Certified Buffered Gluing for the K3 Oka Manuscript

Residual Balls, Inexact Corrections, and Domain-Sensitive Error Budgets

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Abstract

We study the analytic interfaces in the manuscript *Every complex K3 surface is Oka*, separating its all-surface geometric claims from quantitative statements about specified holomorphic transition maps. We derive an a posteriori residual-ball certificate for planar spray gluing, an explicit nonlinear step estimate with an imperfect but holomorphic additive splitting, and a geometrically convergent inexact iteration with finite stopping and side-dependent target error bounds. Exact quadratic majorants allow a finite prefix of individually certified splitting constants, while a weighted-product calculation confirms that the source's dyadic margin schedule is already optimal for the homogeneous scalar model. Annular Fourier truncations are treated without discarding symplectic flux or the positive transverse radius. The conclusions are conditional on certified domains, holomorphy, and operator bounds; neither finite tests nor local moment coordinates establish the global K3 theorem. Contribution assessment: C3/5, the explicit inexact and a posteriori gluing certificates provide notable source-specific quantitative control, without a new global Oka theorem or a historical novelty claim.

1 Question, Status, and Logical Boundary

The source [1] asserts that for every complex K3 surface X, every integer $m \geq 1$, every nonempty compact convex $K \subset \mathbb{C}^m$, every map f holomorphic near K, and every $\varepsilon > 0$, an entire $F : \mathbb{C}^m \to X$ satisfies

$$\sup_{z \in K} d_X(F(z), f(z)) < \varepsilon.$$

The metric is any smooth Hermitian distance on the compact target. This is CAP in every source dimension, not merely dominability, vanishing Kobayashi pseudodistance, or approximation by entire curves. Forstnerič's CAP characterization is a classical external input [2]. This report does not independently certify the asserted global theorem.

Our local problem is narrower and completely specified: given holomorphic sprays on two buffered pieces, turn a verified small transition error into an exact glued central map and a quantitative approximation estimate. A verified bound is in the stated function norm on a stated domain, not a small list of point samples. The local theorems below apply to arbitrary finite fibre dimension. Their use for K3 surfaces retains all the source's geometric obligations.

The public comparison was refreshed on October 8, 2026. Xie and Zhao's September 17, 2026 version of *Periods of Oka K3 surfaces are dense* proves density statements and Oka results for

Wehler Calabi–Yau manifolds, including the $(2,2,2)$ K3 case [4]. Its introduction still states the all-K3 assertion as a conjecture. Density is not a proof for every period. Classical nonlinear splitting, including parameter-dependent versions, already supplies the broad mechanism used here [3]. Our contribution is the explicit certification and error allocation at the source's particular interfaces, not the invention of the splitting method.

1.1 The gates that must be retained

A map on a compact set is understood to be holomorphic on an open neighborhood. All fibre balls use the maximum norm on $\mathbb{C}^N$. On an open set Ω, $H^\infty(\Omega, \mathbb{C}^N)$ has the supremum norm. Norms of derivatives are induced operator norms, not entrywise maxima. Where a vector of operators is used, its norm is stated separately.

The source's property L is local in the *target*: a witness neighborhood V permits every planar special pair $D \subset D'$, every passive polydisc pair $P_0 \Subset \operatorname{int} P$, and every map near $D \times P$ with image in V, to be approximated on $D \times P_0$ by maps near $D' \times P_0$. No uniformity over all such inputs is implicit. The following data are fixed before choosing the approximation accuracy:

1. the base overlap geometry, cutoffs, and positive base margins;
2. a positive common auxiliary fibre radius and inverse-chart margins;
3. the finite list of derivative orders used by the correction;
4. a compact set on each side where a target continuity modulus is available, if approximation on that side is claimed.

In `gluing.tex`, the polynomial change of coordinates and its cutoff constants are fixed before invoking arbitrarily accurate planar approximation. Arbitrary accuracy for a sequence whose fibre radii collapse is not the same hypothesis.

2 A Planar Residual-Ball Certificate

Let $U_A, U_B \subset \mathbb{C}$ be bounded open sets with positively separated closed fringes, $C = U_A \cap U_B$, and P an open passive polydisc. The source constructs bounded linear operators

$$R_A : H^\infty(C \times P, \mathbb{C}^N) \to H^\infty(U_A \times P, \mathbb{C}^N), \qquad R_B : H^\infty(C \times P, \mathbb{C}^N) \to H^\infty(U_B \times P, \mathbb{C}^N)$$

with $R_A u - R_B u = u$. Write $\|R_A\| \le \kappa_A$ and $\|R_B\| \le \kappa_B$. These constants are independent of the number and radii of passive parameters for this planar construction. This independence is not asserted for the multidimensional splitting.

For completeness, with a cutoff χ zero on the A-only fringe and one on the B-only fringe, the Cauchy–Green solution is

$$v(s,t) = \frac{1}{\pi} \int_C \frac{u(\zeta,t)\, \partial_{\bar\zeta} \chi(\zeta)}{s - \zeta}\, dA(\zeta), \qquad R_A u = \chi u - v, \quad R_B u = (\chi - 1)u - v.$$

If d is the diameter of $U_A \cup U_B$ and $0 \le \chi \le 1$, each operator has norm at most $1 + 2d \left\| \bar\partial \chi \right\|_\infty$. Indeed the integral of $|s-\zeta|^{-1}$ over a containing radius-d disc is $2\pi d$. The distributional fundamental-solution identity proves holomorphy after subtraction. Dominated integration preserves holomorphic passive parameters. This is the source's bounded splitting, not a new representation theorem.

Let $\gamma(z,v) = v + E(z,v)$ be the transition with the convention that gluing asks for $a = \gamma(z,b)$. Substituting $a = R_A u$, $b = R_B u$ gives the exact fixed-point equation

$$u = T(u) := E(z, R_B u(z)). \tag{1}$$

This reformulation uses an exact right inverse of the jump map. A rectangular least-squares residual would not supply that identity.

Theorem 2.1 (Certified correction about an approximate solution)**.** *Let $\widehat{u} \in H^\infty(C \times P, \mathbb{C}^N)$ and $r > 0$. Suppose γ is holomorphic for $\|v\| < b$, its derivative satisfies $\|D_v E\| \le L$ on a convex fibre set containing all $R_B u$ for $\|u - \widehat{u}\| \le r$, and all those evaluations have a positive domain margin. Set*

$$\eta = \|\widehat{u} - E(\,\cdot\,, R_B\widehat{u})\|\,, \qquad q = L\kappa_B < 1, \qquad \eta + qr \le r.$$

Assume also that both side sprays are defined at $R_A u, R_B u$ for every u in that ball. Then there is a unique solution u_* of (1) in that ball, and

$$\|u_* - \widehat{u}\| \le \frac{\eta}{1-q}, \qquad \|R_\sigma u_* - R_\sigma \widehat{u}\| \le \frac{\kappa_\sigma \eta}{1-q} \quad (\sigma = A, B). \tag{2}$$

For a known target modulus ω_σ of the fixed side spray, the target discrepancy from the approximately corrected central map is at most $\omega_\sigma(\kappa_\sigma\eta/(1-q))$.

Proof. The derivative bound and integration along fibre segments give $\|T(u) - T(v)\| \le q\,\|u - v\|$. Also $\|T(u) - \widehat{u}\| \le qr + \eta \le r$. The closed ball in H^∞ is complete, so contraction gives the unique fixed point. Comparing it to $\widehat{u}$ gives $(1-q)\,\|u_* - \widehat{u}\| \le \eta$. Bounded splitting gives the side bounds, and applying the stated modulus gives the target estimate. Equation (1) and $R_A - R_B = I$ prove exact equality of the corrected maps. □

This is a direct adaptation of the residual-ball and composition arguments in the original private nonlinear proof text [6], freshly read for this report. It improves the source's qualitative smallness test by accepting a computed holomorphic centre $\widehat{u}$ and unequal side constants. It does not improve the underlying Banach fixed-point theorem. Finite samples cannot replace η or the derivative supremum.

3 Imperfect Additive Splitting, Exact Holomorphic Gluing

The buffered multidimensional construction instead iterates

$$\gamma_{\text{new}} = (\mathrm{id} + b)^{-1} \circ (\mathrm{id} + c) \circ (\mathrm{id} + a).$$

The following elementary estimate exposes the error term that an implemented additive split must certify.

Proposition 3.1 (One step with a splitting residual)**.** *Fix $\rho, h > 0$. On a base overlap let a, b, c be jointly holomorphic in the base and $v \in B_{\rho+2h}$. Assume there are nonnegative bounds*

$$\|a\| \le \alpha \le h, \quad \|D_v c\| \le L_c, \quad \|D_v b\| \le \beta < 1, \quad \|c + a - b\| \le R.$$

Put $D = (R + L_c\alpha)/(1-\beta)$ and suppose $D \le h$. Then for $v \in B_\rho$ there is a unique holomorphic d with $\|d\| \le D$ satisfying

$$(\mathrm{id} + b)(v + d) = (\mathrm{id} + c)(v + a).$$

In particular, on these images the inverse expression above is well-defined and its error is bounded by D.

Proof. Let $r(v) = c(v) + a(v) - b(v)$. The equation is exactly

$$d = r(v) + c(v + a) - c(v) - \{b(v + d) - b(v)\}. \tag{3}$$

The right-hand side maps the closed D-ball to itself and has Lipschitz constant at most β in d. The line segments used lie inside $B_{\rho+2h}$. Contraction, uniformly on compact subsets of the base and fibre, produces a holomorphic solution. The derivative bound on b also gives $\|(v + b(v)) - (w + b(w))\| \geq (1 - \beta)\|v - w\|$ on that convex ball. Thus the solution is the stated inverse branch, not merely a formal equation. □

For the source's split, $\alpha = O(h^{-q_0}e)$, $L_c = O(h^{-1}e)$ by fibre Cauchy estimates, and β is small after a further fixed margin loss. With $\beta \leq 1/2$, a certified residual R gives

$$e_{\text{new}} \leq Ch^{-p}e^2 + 2R \tag{4}$$

after increasing one fixed exponent p for the finitely many estimates involved. Additional perturbations of the transition must be incorporated in the actual holomorphic residual, not omitted. This yields an inexact version of the source's `eq:glue-quadratic`.

On a real interface the norm is instead

$$\|c\|_\alpha = \sup_{v \in B_b} \|c(\,\cdot\,, v)\|_{C^\alpha(M_r)}, \quad 0 < \alpha < 1, \quad M_r = Q_r \cap \{\operatorname{Im} z_1 = 0\}.$$

Side functions carry both their supremum norm and their C^α trace norm. To apply (3) there, the derivative and composition bounds must hold in this Banach algebra: the second fibre derivative controls the change of $D_v c$ at shifted arguments. The source establishes such bounds with finite powers of h^{-1}. A pointwise derivative bound alone does not certify the trace step. Smallness on a shrinking complex collar is not substituted for the trace norm.

4 A Robust Inexact Iteration

Fix $h_i = h_0 2^{-i}$, $h_0 > 0$. All intermediate losses within one step are included in a fixed multiplier c_0 of h_i; therefore the total reserved margin is $2c_0h_0$.

Theorem 4.1 (Geometric accuracy is sufficient)**.** *Let $A > 0$, $p \geq 0$, $q \geq 0$, and $s > \max(p, q)$. Suppose actual holomorphic correction steps satisfy all their inverse and domain gates and*

$$e_{i+1} \leq A2^{pi}e_i^2 + \delta_i, \qquad e_i, \delta_i \geq 0.$$

Choose

$$0 < E \leq \frac{2^{-s-1}}{A}, \qquad e_0 \leq E, \qquad \delta_i \leq \frac{E}{2}\,2^{-s(i+1)}. \tag{5}$$

Then $e_i \leq E2^{-si}$ for every $i \geq 0$. For every fixed loss exponent $0 \leq q < s$ and $n \geq 0$,

$$\sum_{i=n}^{\infty} h_i^{-q} e_i \leq h_0^{-q} E \frac{2^{-(s-q)n}}{1 - 2^{q-s}}. \tag{6}$$

Proof. Assume $e_i \leq E2^{-si}$. Dividing the quadratic term by $E2^{-s(i+1)}/2$ gives $2AE2^s2^{(p-s)i} \leq 1$. Thus the quadratic and inexact terms each consume at most half the next bound. This proves the induction, including its initial step. Summing the resulting geometric series proves (6). □

The improvement is a tolerance policy: truncation errors need not decrease doubly exponentially to obtain exact limiting agreement. The theorem does not permit an arbitrary nonholomorphic approximation, nor does it establish the gates from its scalar recurrence. For example, if a gate is $D_0 h_i^{-g} e_i \le h_i/4$, it is sufficient in addition to require

$$s \ge g+1, \qquad E \le h_0^{g+1}/(4D_0). \tag{7}$$

All finitely many gates must be checked this way or individually. The weaker condition $s > q$ alone only proves summability. A fixed positive allowed error does not guarantee exact gluing: the recurrence with $\delta_i = \delta > 0$ permits $e_i = \delta$ after the first step and hence no vanishing defect.

5 Actual Constants and the Optimal Homogeneous Schedule

Proposition 5.1 (Unequal certified constants)**.** *If $e_{i+1} \le A_i e_i^2$ with $A_i > 0$, then*

$$e_n \le e_0^{2^n} \prod_{i=0}^{n-1} A_i^{2^{n-1-i}}. \tag{8}$$

If $\sum_{i\ge 0} 2^{-i-1} |\log A_i| < \infty$ and $G = \prod_{i\ge 0} A_i^{2^{-i-1}}$, the condition $e_0 G < 1$ implies eventual doubly exponential decay and convergence of $\sum_i 2^{qi} e_i$ for every fixed $q \ge 0$.

More usefully for a finite certificate, suppose the first n steps give a certified bound $\bar{e}_n$ and $A_i \le \bar{A} 2^{pi}$ for $i \ge n$, where $\bar{A} > 0$, $p \ge 0$. If $\theta_n = \bar{A} 2^{p(n+1)} \bar{e}_n < 1$, then

$$e_{n+j} \le \bar{A}^{-1} 2^{-p(n+j+1)} \theta_n^{2^j} \quad (j \ge 0). \tag{9}$$

Proof. Induction gives (8). Its right-hand side equals $(e_0 G_n)^{2^n}$, where $G_n = \prod_{i<n} A_i^{2^{-i-1}} \to G$. Choose $e_0 G < \theta < 1$; eventually $e_0 G_n \le \theta$. The weighted series then converge by comparison with $2^{qi}\theta^{2^i}$. For the finite-prefix assertion, set $x_j = \bar{A} 2^{p(n+j+1)} e_{n+j}$. Direct substitution gives $x_{j+1} \le x_j^2$, proving the formula. □

This retains actually certified favorable early constants instead of replacing them all by the worst envelope. It does not estimate any unknown splitting constant. For example, take $p = 2$, $\bar{A} = 16$, $A_0 = 1$, and $A_i \le 16\, 2^{2i}$ for $i \ge 1$. At $e_0 = 1/32$, the all-step envelope test fails because $16 \cdot 4e_0 = 2$. The actual first step gives $e_1 \le 1/1024$ and $\theta_1 = 16 \cdot 16/1024 = 1/4$, so the exact tail passes. This is a scalar certificate fixture, not measured data from a K3 atlas. All domain gates remain necessary.

Proposition 5.2 (No gain from redistributing homogeneous losses)**.** *Fix $C, H, p > 0$ and positive losses h_i with $\sum_i h_i \le H$. For $A_i = C h_i^{-p}$, the logarithmic majorant satisfies*

$$\prod_{i\ge 0} A_i^{2^{-i-1}} \ge C(4/H)^p.$$

Equality holds exactly for $h_i = H 2^{-i-1}$. If the weighted logarithms diverge, the left side is interpreted as $+\infty$.

Proof. Put $w_i = 2^{-i-1}$; then $\sum_i w_i = 1$ and $\sum_i (i+1) w_i = 2$. Jensen's inequality for log gives

$$\sum_i w_i \log \frac{h_i}{H w_i} \le \log \frac{\sum_i h_i}{H} \le 0.$$

The positive part is integrable since $\log^+ t \le t$; thus the inequality is valid also when the negative part diverges, by truncation or the extended form of Jensen's inequality. As $\sum_i w_i \log w_i = -2\log 2$, exponentiation proves the bound. Strict concavity gives equality only when $h_i/(Hw_i)$ is constant, and the budget must then be saturated. □

For $h_i = H(1-r)r^i$, $0 < r < 1$, the reciprocal scalar threshold is $CH^{-p}(r(1-r))^{-p}$, minimized at $r = 1/2$. Thus the source's dyadic choice is optimal not only among geometric schedules but among all schedules in this *homogeneous model*. This fact is a classical weighted-product observation, not credited as a new algorithmic gain. Different loss exponents, domain-dependent constants, or additional gates change the optimization problem.

6 Composition Tails and a Finite Stopping Rule

The estimate for transition errors is not itself an estimate for the accumulated coordinate maps. The order of composition matters.

Theorem 6.1 (Nested side maps)**.** *For $\sigma = A, B$, let $\psi_i^\sigma = \mathrm{id} + a_i^\sigma$ be holomorphic fibre maps over fixed base points, and put $C_0^\sigma = \mathrm{id}$, $C_{i+1}^\sigma = C_i^\sigma \circ \psi_i^\sigma$. Assume nested domain inclusions keep all compositions and the fibre segments in this proof inside their prescribed buffered domains. Suppose, on these domains,*

$$\|a_i^\sigma\| \le d_i^\sigma, \qquad \|D_v a_i^\sigma\| \le l_i^\sigma, \qquad \sum_i d_i^\sigma < \infty, \quad \sum_i l_i^\sigma < \infty.$$

Set $L_\sigma = \prod_{i\ge 0}(1 + l_i^\sigma) \le \exp(\sum_i l_i^\sigma)$. On every common retained domain, C_i^σ converges locally uniformly to a holomorphic map C_∞^σ, with

$$\|C_\infty^\sigma - C_n^\sigma\| \le L_\sigma \sum_{i=n}^{\infty} d_i^\sigma. \tag{10}$$

If the actual conjugated transitions tend to the identity, the original side sprays composed with C_∞^A, C_∞^B agree. For a target modulus ω_σ valid on the compact propagated set,

$$d_Y(F_\sigma \circ C_\infty^\sigma, F_\sigma \circ C_n^\sigma) \le \omega_\sigma\left(L_\sigma \sum_{i=n}^{\infty} d_i^\sigma\right). \tag{11}$$

No target estimate on a side lacking such a modulus is asserted.

Proof. The chain rule and the segment hypothesis bound the Lipschitz constant of C_i^σ by $\prod_{j<i}(1+l_j^\sigma)$. Consequently $\|C_{i+1}^\sigma - C_i^\sigma\| \le L_\sigma d_i^\sigma$. Telescoping proves the Cauchy property and the tail bound. Locally uniform limits of holomorphic coordinate maps are holomorphic. The exact identities defining the conjugated transitions pass to the limit on compact retained domains, since their error tends to zero and the original sprays are continuous there. This proves agreement; the modulus gives the last inequality. For trace gluing the same argument uses convergence of the side traces, followed by Morera's theorem in target charts across the real interface, exactly as in the source. □

When $d_i^\sigma \le D_\sigma h_i^{-q_\sigma} e_i$, Theorem 4.1 provides an explicit tail

$$T_{\sigma,n} = L_\sigma D_\sigma h_0^{-q_\sigma} E \frac{2^{-(s-q_\sigma)n}}{1 - 2^{q_\sigma - s}}. \tag{12}$$

A finite stopping rule is $\omega_\sigma(T_{\sigma,n}) \le \varepsilon_\sigma$ on each side where accuracy is required. For a Lipschitz target spray use $\omega_\sigma(t) = M_\sigma t$. One may use the actual finite product through $n-1$ times a certified tail bound for the derivative product, rather than exponentiating all the observed constants. If injectivity of the limit is needed, the additional conditions $l_i^\sigma < 1$ and the corresponding segment hypotheses give the positive lower product $\prod_i (1 - l_i^\sigma) > 0$. Mere convergence alone would not do so.

Example 6.2 (An exact rational budget)**.** Let $p = 2$, $A = 4$, $h_0 = 1/8$, $q = 2$, $s = 5$, and $E = 1/256$. Then $E = 2^{-s-1}/A$, and the maximal allowed defect is $\delta_i = 2^{-5i-14}$. The total weighted error is at most

$$h_0^{-2} E/(1 - 2^{-3}) = 2/7.$$

It is not automatically a small coordinate displacement: a side constant D_σ and its domain margin must still be tested. For a gate $h_i^{-2} e_i \le h_i/4$, reduce E to at most $1/2048$. This also reduces the weighted sum to $1/28$. The explicit distinction prevents acceptance based on a scalar convergence test that violates the actual fibre radius.

7 Holomorphic Moment Truncations and Annular Flux

The private local-moment theorem [7] expands an integral operator by Taylor's formula, with an absolute-moment remainder. Its proof requires all Taylor segments to stay in the function's domain and gives no derivative estimate from a small pointwise remainder alone. These are directly relevant warnings here. For holomorphic gluing one must additionally retain holomorphy in the base and in the fibres. A real-variable moment approximation with a coefficient $\bar{z}$, for example, need not do so even when its pointwise error is small.

Proposition 7.1 (A usable holomorphic coefficient certificate)**.** *Let $f(z, v)$ be jointly holomorphic and bounded by M on $\Omega \times \prod_{j=1}^k \{|v_j| < R_j\}$. Choose $0 < \rho_j < R_j$, set $t_j = \rho_j/R_j$, and retain $0 \le \alpha_j \le N_j$ in the fibre Taylor series. Its coefficients $c_\alpha(z)$ are holomorphic on Ω. If implemented coefficients $\widehat{c}_\alpha$ are holomorphic there and satisfy $\|\widehat{c}_\alpha - c_\alpha\| \le \eta_\alpha$, then on the smaller product*

$$\left\| f - \widehat{f} \right\| \le M \prod_{j=1}^k (1 - t_j)^{-1} \sum_{j=1}^k t_j^{N_j+1} + \sum_{\alpha \text{ retained}} \eta_\alpha \rho^\alpha. \tag{13}$$

On a further fibre loss $\sigma > 0$, each multi-index derivative of this remainder is bounded by $\alpha! \sigma^{-|\alpha|}$ times the displayed bound.

Proof. Cauchy's coefficient formula gives $|c_\alpha| \le M \prod_j R_j^{-\alpha_j}$ and holomorphic dependence on z. Sum the absolute series outside the retained box. Its geometric sum is

$$\prod_j (1 - t_j)^{-1} \left[1 - \prod_j (1 - t_j^{N_j+1}) \right],$$

which is bounded by the first term of (13). The finite coefficient perturbation contributes the second term. The remainder is holomorphic, so Cauchy's derivative estimate on the additional polydisc margin proves the last claim. □

This supplies legitimate input errors for Proposition 3.1 when the polydiscs and coefficients actually cover the required domains. For a first derivative in the maximum norm, the induced operator bound may cost a factor k; it is not just the largest single partial derivative. A real-interface C^α norm must be certified separately, or obtained from stronger base bounds on a genuinely available fixed neighborhood.

7.1 Why a small symplectic defect is different

The annular source uses $w \in \mathbb{C}/\mathbb{Z}$, $p \in \Delta_r$, $\omega = dw \wedge dp$, and the primitive convention $\lambda = p\,dw$. For a degree-one symplectic transition $F = (W, P)$, exactness is the exact identity

$$\int_{\mathbb{R}/\mathbb{Z}} (P\,dW - p\,dw) = 0.$$

The map $(w, p) \mapsto (w, p + \epsilon)$ is holomorphic, symplectic, and arbitrarily close to the identity, but its period is ϵ. It is not exact unless $\epsilon = 0$. Conjugating by exact Hamiltonian changes cannot remove this cohomological obstruction: exact changes add exact one-forms, and a degree-one change preserves the period. The source correctly recentres the newly added chart to cancel the period and separately proves that the moving centres remain in uniformly buffered regions. That step is not an error estimate.

7.2 A certified finite Fourier stencil

The source's mean-zero Hamiltonians H_j are split by forward negative modes and backward positive modes. Its coefficient-line estimate has the following precise numerical form: on a δ-buffered block, each signed contribution at frequency $n \geq 1$ and block distance $q \geq 0$ is bounded by

$$e \exp[-n(a + bq)], \qquad a = 3\pi\delta/2, \quad b = 2\pi(1 - \kappa), \quad 0 \leq \kappa < 1/4.$$

This uses the uniform lower block length $L_j \geq 1$, not an upper bound on chain length.

Proposition 7.2 (Uniform truncation error)**.** *Retain $1 \leq n \leq N$ and $0 \leq q \leq Q$ on each of the two sides of this Fourier split, where N, Q are nonnegative integers. For the resulting holomorphic finite stencil $G_j^{N,Q}$,*

$$\sup_j \left\| G_j - G_j^{N,Q} \right\| \leq \frac{2e}{(1 - e^{-a})(1 - e^{-b})} \left(e^{-a(N+1)} + e^{-a} e^{-b(Q+1)} \right) =: \tau. \tag{14}$$

On a further common seam where both block estimates apply, the additive Hamiltonian jump defect has norm at most 2τ.

Proof. For $n > N$, sum over q first and use $1 - e^{-bn} \geq 1 - e^{-b}$; then sum over n. This gives $e^{-a(N+1)}/((1 - e^{-a})(1 - e^{-b}))$. For $q > Q$, bound $e^{-bnq} \leq e^{-bq}$ and sum the two geometric series. Multiply by $2e$ for the two signed sides. The union bound can count the omitted corner twice, which is harmless. Normal convergence justifies comparison with the exact split. Subtract its exact jump identity from the truncated one: the two tails, one composed with the adjacent shear, give 2τ on the stated common domain. □

Writing $D = (1 - e^{-a})(1 - e^{-b})$, a desired tail $\tau_* > 0$ is ensured by the explicit choices

$$N + 1 \geq a^{-1} \log \frac{4e}{D\tau_*}, \qquad Q + 1 \geq b^{-1} \log \frac{4ee^{-a}}{D\tau_*},$$

rounded up and combined with $N, Q \geq 0$. Cauchy estimates on a further actual margin bound the Hamiltonian vector-field jump defect. The finite stencil must be used through its *actual Hamiltonian time-one flow*, whose existence is checked in the reserved domains. Such a flow remains exactly symplectic and exact because $\mathcal{L}_{X_H}(p\,dw) = d(pH_p - H)$. Replacing that flow by a merely close arbitrary polynomial map would not preserve this conclusion.

If the nonlinear step has separately certified bounds $e_{\mathrm{new}} \le Ch^{-p}e^2 + C'h^{-q}\tau$, choose the Fourier tail so that $C'h^{-q}\tau$ meets (5). This is a conditional implementation route, not a newly proved uniform sewing theorem with unspecified constants. As δ decreases dyadically, a decreases too: the required frequency cutoff can grow on the order of $2^i i$ for exponential relative tolerances, whereas the block-distance cutoff is of order i. Thus this certificate does not promise constant computational cost. Each stencil uses finitely many neighbors, but applying it to every $j \in \mathbb{Z}$ still describes an infinite chain.

8 Source Architecture and Retained Obligations

The cached build comprises thirteen TeX files and one bibliography, 4,425 literal lines in total. All were read in bounded displays. This is full literal coverage of this build, *not* a claim that every deep theorem it invokes has been independently checked. The following map specifies the logical chain.

Module and lines	Role in the source	Boundary in this report
Introduction 1–229; paper 1–68	States all-dimensional CAP for every complex K3; identifies earlier partial flexibility results.	Source claim and quantifiers recorded; not promoted to an independently verified global theorem.
Local 1–553	Complete aligned strips imply L; planar bounded splitting, chart sprays, finite sector filling, propagation across an analytic exceptional locus.	The actual splitting and fixed-point identities are used. Stein neighborhoods, Cartan theory, scalar approximation and Oka–Grauert triviality remain classical inputs.
Gluing 1–549	Buffered open splitting with polynomial margin loss; polynomial automorphism concentration; L with passive parameters gives polydisc approximation.	The transition recurrence is quantified locally. The concentration/enlargement construction is read and mapped, not replaced by the finite certificates.
Globalization 1–646	Complexification of real-parameter paths; face approximation; C^α interface splitting; induction removing real faces.	The trace norm and finite derivative losses are retained. The weighted $\bar\partial$ theorem and interior elliptic estimates are imported analytic inputs.
Projective 1–269	Distinct ample degrees in genus-one families force transverse directions; complete flows use full proper normalized fibres.	Chen–Gounelas curve existence is imported. Restricting inverse sheets does not restrict the full fibres on which flows exist.
Sewing 1–436	Exactness removes the transverse zero mode; two-sided Fourier splitting; Hamiltonian quadratic iteration; recentering.	The finite stencil is proved here from the displayed coefficient bound. Uniform geometric seam construction and exact period adjustment are not finite-test conclusions.
Degeneration 1–561	Two quadrics, sixteen blowup centres, regular pencils, core/neck charts, compact moving-centre coverage, unrestricted deformation.	The finite raw atlas must retain widths, inverse branches, coverage and transverse partners. These are substantive source arguments, not consequences of a scalar recurrence.
Rational periods 1–543	An isotrivial centre for each primitive positive integral v; safe symplectic actions; uniform strips; two independent cycles.	The normalization $[\mathrm{Im}\,\sigma] = v$, exact action conservation, and compact avoidance of singular fibres are retained. Nearby fibrations are not assumed to persist.

Module and lines	Role in the source	Boundary in this report
Dynamics 1–171	Zero, one, or two rational directions; Ratner/Borel orbit alternatives; Torelli transfer.	Deep arithmetic dynamics and K3 period theory are imported. Density alone is insufficient; the one-direction case needs an open set inside its fixed-v locus.
Consequences 1–164	Oka-1/interpolation, dense immersions, Enriques and minimal-surface consequences, projective global sprays.	These depend on the source's global theorem. Class VII additionally uses a separate companion global-spherical-shell assertion, not verified here and not needed for the K3 argument.
Figures 1–30 and 1–25; bibliography 1–181	Planar pieces, real interface, and reference metadata.	Read as part of the build; reading bibliography entries does not amount to reading the referenced works.

8.1 What was checked, and what was not

For the analytic steps used here, no correction to the source's displayed quadratic identity or dyadic rescaling was needed. The source also correctly keeps a fixed positive fibre ball before requesting arbitrary accuracy, retains the finitely many derivatives needed at each step, and states approximation only on a side with a controlled target modulus. Its real-face construction does not assume an unjustified fixed normal collar. Its annular construction separates exact period cancellation from smallness. These valid details are retained rather than treated as new repairs.

The audit makes no claim of a global source-proof certification. In particular, the available local constants are principally existence bounds, not numerically evaluated constants for a specified atlas. The unbounded-chain coverage and recentering arguments, the whole deformation transfer, and the global orbit classification have been read as dependencies; they have not been recreated as independent global proofs. No numerical experiment in this report constructs an Oka map on a K3 surface.

For the confidential original corpus, the literal reads were `stability.tex` lines 1–295 and `ch03-local-moments-rev2.tex` lines 1–160 and 227–313: 542 of these two files' 741 lines. The latter file's unread complements are 161–226 and 314–446. Its separately included chapters were not implicitly read. The private material remains local; the references below identify the proof roles, not public downloads. Pre-read and final hashes, merged intervals and complements are recorded in the owned read ledger. Equal endpoint hashes establish the observed endpoints, not continuous monitoring or authentication of a live repository branch.

9 Sharp Limits, Comparisons, and Further Work

Small scalar residual versus a holomorphic domain. Even if $\|c\|_j \to 0$ on B_{b_j}, the ratio $\|c\|_j / b_j$ need not decrease. For $b_j = \varepsilon_j^2$ and $c_j(v) = \varepsilon_j$, it diverges as $\varepsilon_j \to 0$. A correction of size comparable to ε_j cannot be assumed to fit a radius ε_j^2. Similarly, $\sum_i h_i^{-q} e_i < \infty$ does not imply $h_i^{-q} e_i = o(h_i)$: set $e_i = 2^{-si}$ with $q < s < q+1$. These are explicit reasons for the separate pointwise gates.

Trace accuracy versus collar accuracy. For each fixed integer $k \geq 0$, the entire functions $f_n(z) = n^{-k-2} \sin(nz)$ tend to zero in $C^k([-1,1])$, and in C^α when $k \geq 1$. For every fixed $y > 0$, however, $|f_n(iy)| = n^{-k-2} \sinh(ny) \to \infty$. Thus tangential accuracy does not provide a uniform

complex thickness. The source's trace operator solves a real analytic interface problem that ordinary buffered sup-norm gluing alone does not solve.

Advantages. The residual ball uses an actual computed centre, asymmetric side norms, and an explicit error-to-target conversion. The inexact iteration tolerates decreasing holomorphic truncation errors while converging to exact agreement; finite-tail bounds specify a stopping rule. The prefix test retains favorable early constants. The Fourier stencil gives a quantitative local cost for the chain operator without imposing a bound on block lengths.

Costs and disadvantages. The continuous norm, holomorphy and domain certificates may be more expensive than computing a candidate. Finite point samples do not bound H^∞ errors, and finite moment rank does not establish holomorphic surjectivity or a global Oka property. A small fibre radius makes Cauchy losses large. Trace norms require additional base regularity. Fourier cutoffs increase as margins shrink. Exact Hamiltonian flow evaluation and exact flux preservation are stronger requirements than a small floating-point residual.

Feasible extensions. A useful next step is a certified atlas-specific implementation: interval bounds for the finitely many chart derivatives, a holomorphic coefficient approximation with rigorous tails, and separate lower bounds for inverse margins. For a real interface, such an implementation should certify the C^α algebra estimates directly rather than sample a thin collar. For annular sewing, one should couple finite Hamiltonian stencils with verified flow bounds and exact period normalization. Neither project is claimed to be completed here. The margin optimization also suggests studying heterogeneous loss exponents, but changing a scalar schedule is not evidence for better geometric radii.

10 Verification and Contribution Assessment

The accompanying executable uses exact rational arithmetic for the affine one-step identities, residual-ball fixtures, quadratic prefix products, geometric inexact budgets, finite weighted-product inequalities, composition telescoping, and rational geometric-series versions of the Fourier and Taylor tail bounds. The test JSON reports cases and assertions separately. They check the local algebra and selected boundary fixtures, not the infinitely many analytic hypotheses of a theorem. They neither measure K3 chart constants nor independently test the deep global geometry. Independent review counts are not included in the author's test counts.

The demonstrated result is a source-specific quantitative package: Theorems 2.1, 4.1, and 6.1, with Propositions 3.1, 5.1, 7.1, and 7.2. The weighted schedule result records an existing optimum and receives no improvement credit. Classical contraction, Cauchy estimates, geometric sums and Hamiltonian exactness are attributed as such. C3 reflects the notable local extension to certified inexact gluing and executable finite stopping, not a claim to have originated these general analytic principles. Applicability L3 is a separate classification and is not used to inflate the contribution assessment.

References

[1] OpenAI, *Every complex K3 surface is Oka*, September 23, 2026. Cached source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public manuscript. Source assertions are distinguished from local results throughout.

[2] F. Forstnerič, *Runge approximation on convex sets implies the Oka property*, Ann. of Math. 163 (2006), 689–707. Primary journal record. The CAP characterization is imported, not reproved.

[3] A. Lewandowski, *Splitting Lemma for Biholomorphic Mappings with Smooth Dependence on Parameters*, J. Geom. Anal. 31 (2021), 5783–5798, published online September 23, 2020. Primary article, Theorems 1.1–1.3 and introductory discussion.

[4] S.-Y. Xie and S. Zhao, *Periods of Oka K3 surfaces are dense*, arXiv:2608.24392v2, September 17, 2026. Introduction, Theorems A–E and stated conjecture; not used as a proof of the all-K3 assertion.

[5] A. Alarcón and F. Forstnerič, *Oka-1 manifolds*, Math. Z. 311 (2025), Paper 33. arXiv:2303.15855v5. The primary abstract and journal metadata were checked; its full proof is not audited here.

[6] M. Subbarao, private original nonlinear integral-equation proof corpus, *Perturbation, regularization, and a posteriori control; nonlinear certified neighborhoods; composition*, `TR1-lin-nonlin/stability.tex`, lines 1–295. Confidential local source; relevant proofs are the residual-ball, bounded-coordinate transfer, and composition propositions.

[7] M. Subbarao, private *Local Moments and Ram Differential Operators*, RamCalBooks, volume 1, revision 2, chapter 3, lines 1–160 and 227–313. The Taylor remainder and differentiated product proofs were read; included files and unread complements are outside the declared scope.